

CODE SPORT



In this month's column, we feature a bunch of interview questions in computer science.

Since quite a few of our student readers have requested me to feature more computer science interview questions, let's continue our discussion from last month, with a few more interview questions.

1. You are given two strings, S1 and S2. You are asked to transform String S1 into String S2 by means of the following two operations, namely, insertion and deletion of a character. Each insertion will incur a penalty of 1 and each deletion will incur a penalty of 2. Can you find the minimum cost needed to transform S1 into S2 and also the set of transformations which result in this minimum cost?
2. Consider the same problem discussed in Question (1). Given two Strings S1 and S2, and also the minimum cost needed to transform S1 into S2, can you find the set of transformations, namely the insertion and deletion, which will result in transforming S1 into S2 with the above specified minimum cost? Is this solution unique? Or can there be multiple sets of such transformations which can result in transforming S1 into S2 with the same minimum cost?
3. You are given a large piece of text T and a small string s1 whose length is N. You are asked to find out the closest approximate match to s1 which is present in T. The closest approximate match to S1 is the string s2, which can be transformed into s1 with minimum cost by means of inserting a new character or deleting a character at each of the different positions of the string. Can you write a C program to find the closest approximate match given the text T and string S1? What is the complexity of your code?
4. You are given a large piece of text T of length 'm' and a small string s1 whose length is 'n'.

- You are asked to find out whether S1 is NOT a sub-string of T. (Note that the question is not to find out if S1 is a sub-string, but to find out if S1 is NOT a sub-string.) Can you write a C function 'IsNotASubString' which can perform this task? What is the complexity of your code? If you use the naïve sub-string comparison function, we know that the complexity would be of the order of $O(mn)$. How can you reduce this complexity?
5. Given an integer k, find all its factors. What is the time complexity of your code?
 6. Given a set S of integers from 1 to N where N can be quite large, you are asked to find the set of factors for each integer in the set S. In this case, if you reused the code that you wrote for Question (5), what would be the time complexity of your program? Is it possible to reduce the time complexity of your program?
 7. Given a set S of integers from 1 to N where N can be quite large, you are asked to find the set of prime numbers in the given set S. Would you be able to re-use the code you wrote for Questions (6) and (7)? If so, how can it be done? What is the time complexity of your solution?
 8. Given two arrays A and B of integers where the array lengths are unequal (A is of size m and B is of size n), you are asked to write a program which can compute the following: A-B, A Set Union B (AUB), and A intersection B. What is the complexity of your code?

You are given a set S of N random strings <s1, s2, ... Sn>. Some of the strings are anagrams of each other. You are asked to write a function which can group the anagrams together into sub-groups. The non-anagrams are each placed in their own sub-groups. For example, given the set S: <"now", "Star", "src", "new", "car", "cat", "tac", "rac", "won", "rats", "not">.